

JKBOSE 2026

CLASS 11TH

**TRIGONOMETRIC
FUNCTIONS**

MATHS

LECTURE-01

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TOPICS TO BE COVERED

1

Angle and Measurement of angles

2

[REDACTED]

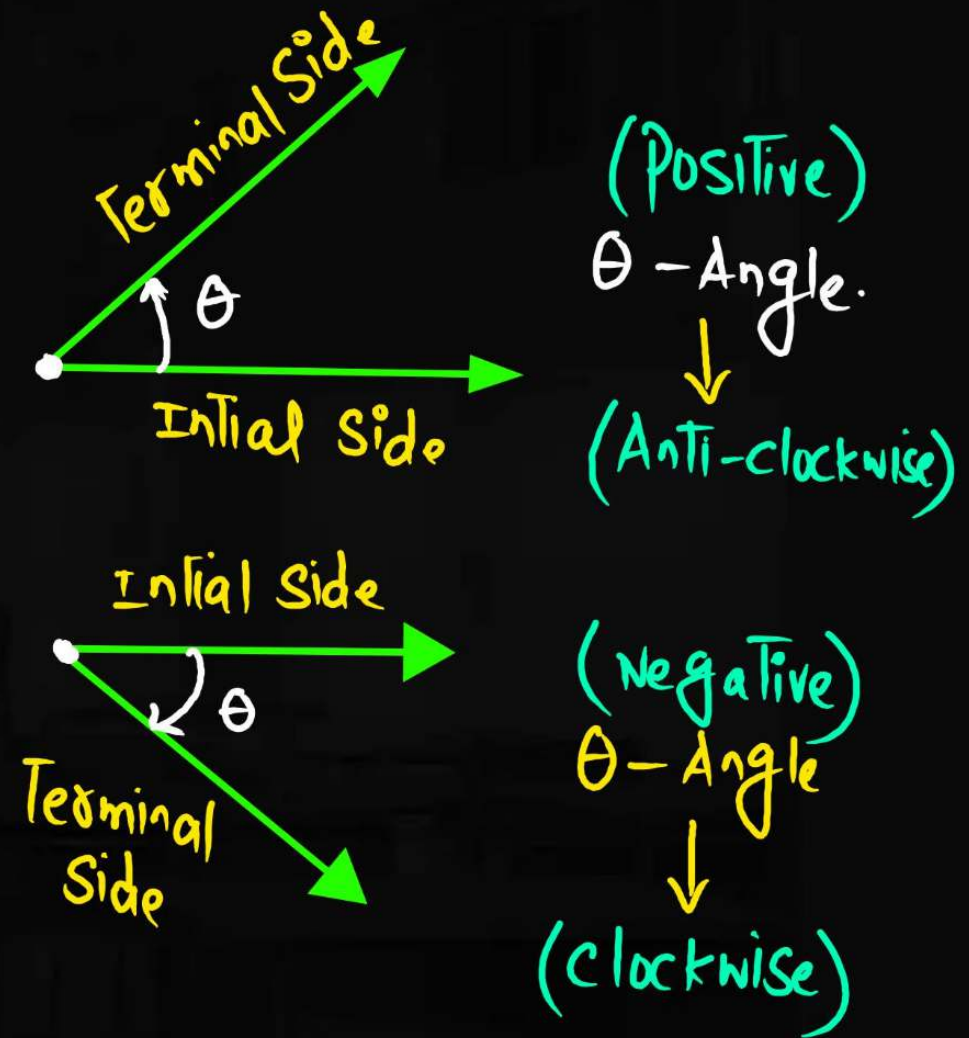
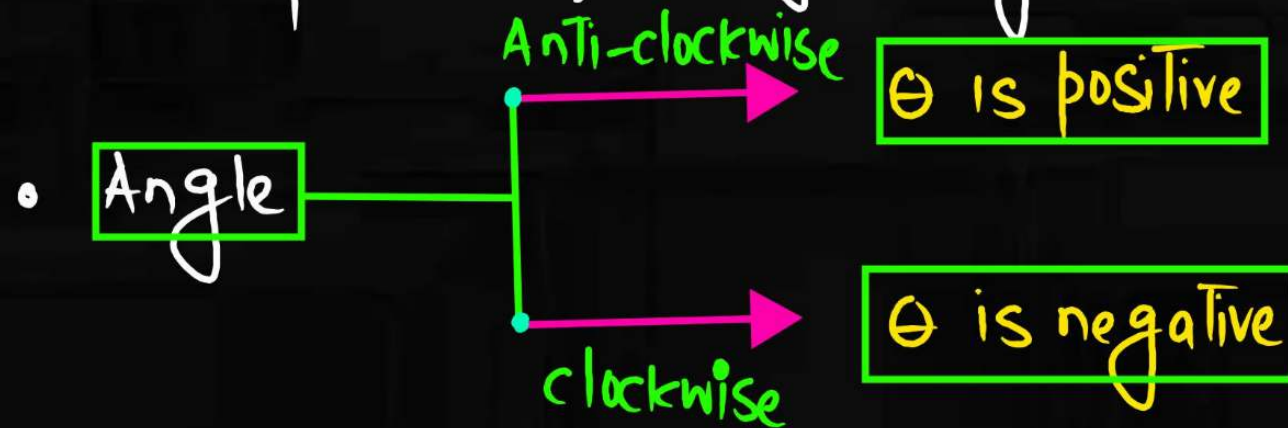
3

Related Problems

Introduction

Angle:

- Amount of rotation of the given ray about its Initial point.
- Initial position of the given ray (Initial Side/ray)
- Final position of the given ray (Terminal Side/ray)



MEASUREMENT OF ANGLES

✓
Sexagesimal system

(British System)

* In this system, we measure an angle in terms of Degrees, minutes & seconds

✗
Centesimal system

(French System)

✓
Circular System

(International System)

* In this system, we measure an angle in terms of Radians

* Right angle:

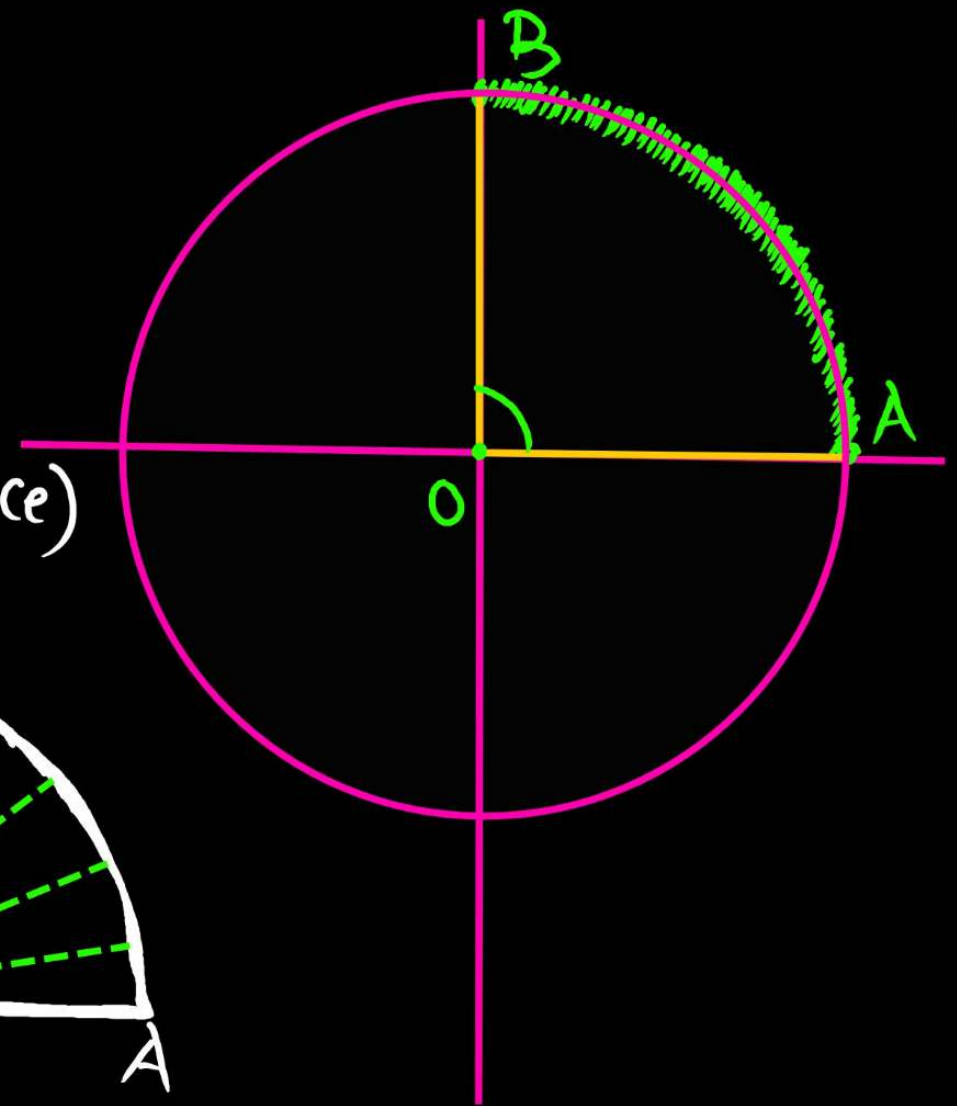
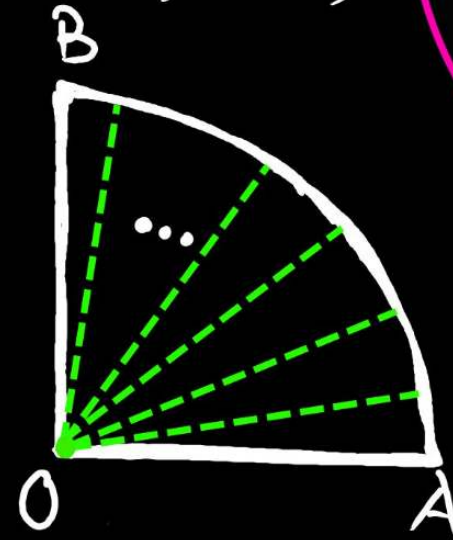
- Angle subtended by the arc AB at the centre of the circle, where $m(\widehat{AB}) = \frac{1}{4}(\text{circumference})$

- 1 Degree is $(\frac{1}{90})^{\text{th}}$ of the right angle.

- Denoted by 1° .

- So, $1^\circ = \frac{1}{90} \times (\text{right angle})$

$$\Rightarrow \boxed{\text{right angle} = 90^\circ}$$



- 1 minute ^{is} $\left(\frac{1}{60}\right)^{\text{th}}$ of 1°

- Denoted by $1'$.

- So, $1' = \frac{1}{60} \times 1^\circ$

$$\Rightarrow 1^\circ = 60'$$

- 1 second is $\left(\frac{1}{60}\right)^{\text{th}}$ of $1'$

- Denoted by $1''$.

- So, $1'' = \frac{1}{60} \times 1'$

$$\Rightarrow 1' = 60''$$

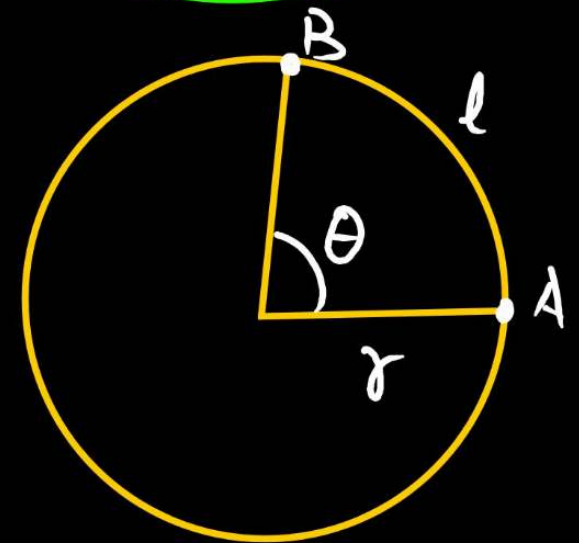
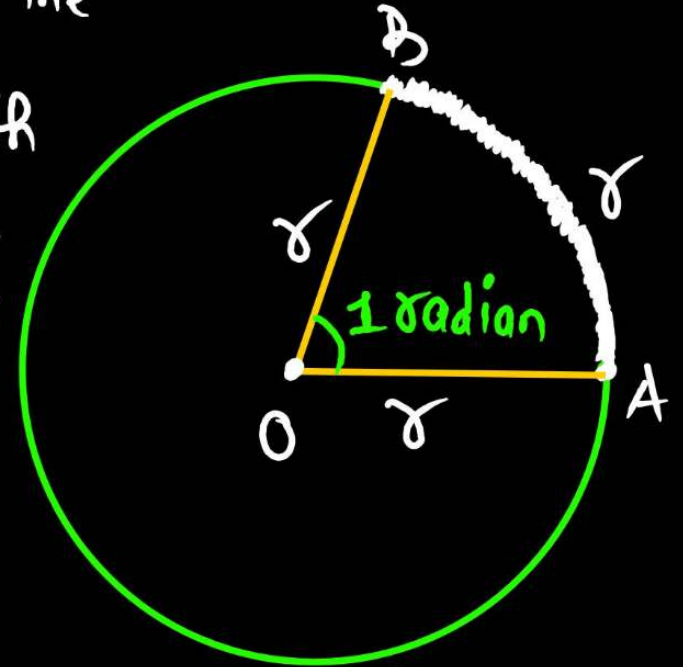
* Radian:

- Angle subtended by an arc at the centre of a circle whose length is same as that of the radius of the circle.

- Here $\theta = \frac{l}{r}$, if $l = r$

$$\Rightarrow \theta = \frac{r}{r} = 1 \text{ radian}$$

$$\Rightarrow \theta = 1 \text{ radian}$$



Conversion between Radians and Degrees

We know that,

$$l = \frac{2\pi r}{2}$$

$$\Rightarrow l = \pi r$$

Also, $\theta = \frac{l}{r} = \frac{\pi r}{r}$

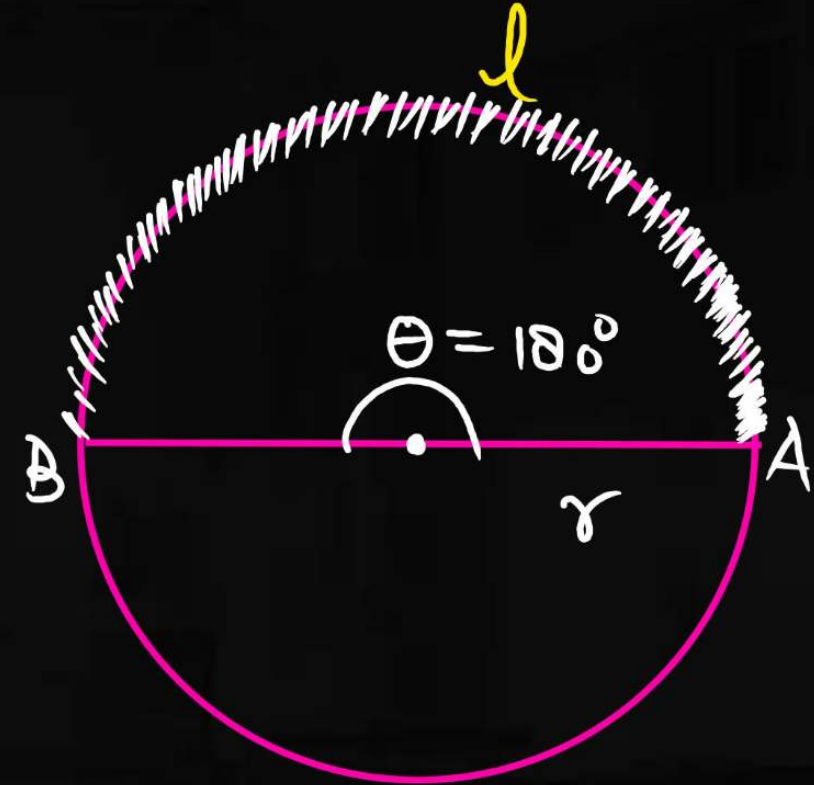
$$\Rightarrow 180^\circ = \pi \text{ radians.}$$

$$\text{So, } 1^\circ = \frac{\pi}{180} \text{ radians}$$

$$\text{and } 1 \text{ radian} = \left(\frac{180}{\pi}\right)^\circ$$

e.g: $90^\circ = 90 \times \frac{\pi}{180}$

$$\Rightarrow 90^\circ = \frac{\pi}{2} \text{ radian}$$



EXAMPLE

Convert $40^\circ 20'$ into radian measure.

(Homework
problem)

QUESTION

Find the degree measure of $\frac{\pi}{9}$ and $-\frac{\pi}{3}$ respectively.

☒ 1 $20^\circ, -60^\circ$

☐ 2 $30^\circ, -60^\circ$

☐ 3 $20^\circ, 40^\circ$

☐ 4 $40^\circ, 50^\circ$

$$\frac{\pi}{9} = \frac{\pi}{9} \times \frac{180}{\pi} = 20^\circ$$

$$-\frac{\pi}{3} = -\frac{\pi}{3} \times \frac{180}{\pi} = -60^\circ$$

QUESTION

Degree measure of an angle of π radians is

☒ 1

90°

☒ 2

180°

☒ 3

20°

☒ 4

50°

$$\pi = \cancel{\pi} \times \frac{180}{\cancel{\pi}} = 180^\circ$$

QUESTION

If the length of a chord of a circle is equal to that of the radius of the circle, then the angle subtended, in radians, at the centre of the circle by the chord is :

~~1~~

1

~~2~~

$\frac{\pi}{2}$

~~3~~

$\frac{\pi}{3}$

~~4~~

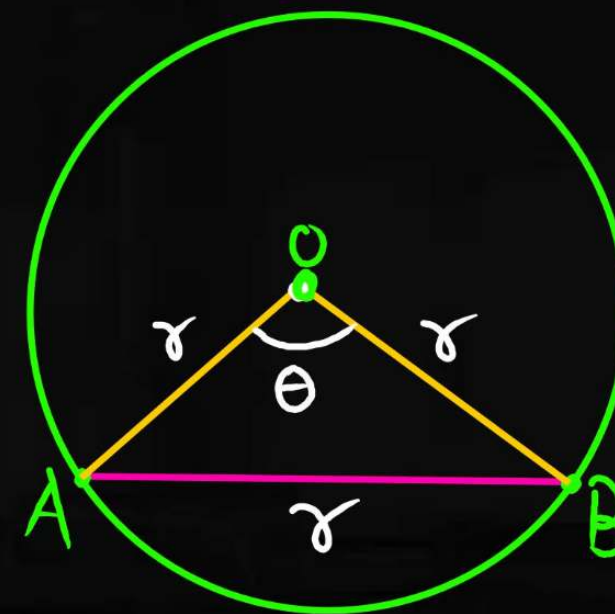
$\frac{\pi}{4}$

(Sol) Here $\triangle AOB$ is an equilateral Triangle.

$$\text{So, } \theta = 60^\circ$$

$$\Rightarrow \theta = 60 \times \frac{\pi}{180}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$



QUESTION

The angle subtended at the centre of circle of radius 3 meters by an arc of 1 length meter is equal to

~~1~~

20°

~~2~~

60°

☒ 3 $\frac{1}{3}$ radian~~4~~

3 radian

$$\theta = \frac{l}{r} = \frac{1}{3}$$

$$\theta = \frac{1}{3} \text{ radians.}$$

QUESTION

The length of an arc of a circle of **radius r** which subtends an angle of **1 radian** at the centre of the circle is

☒ **1** r

☐ **2** $\frac{r}{2}$

☐ **3** $\frac{\pi r}{2}$

☐ **4** $\frac{2r}{\pi}$

Example

If the arcs of the same lengths in two circles subtend angles 65° and 110° at the centre, find the ratio of their radii.

(Sol) Here $\theta_1 = 65^\circ$

$$\Rightarrow \theta_1 = \frac{13}{180} \times \pi \text{ radians}$$

$$\Rightarrow \theta_1 = \frac{13\pi}{36} \text{ radians}$$

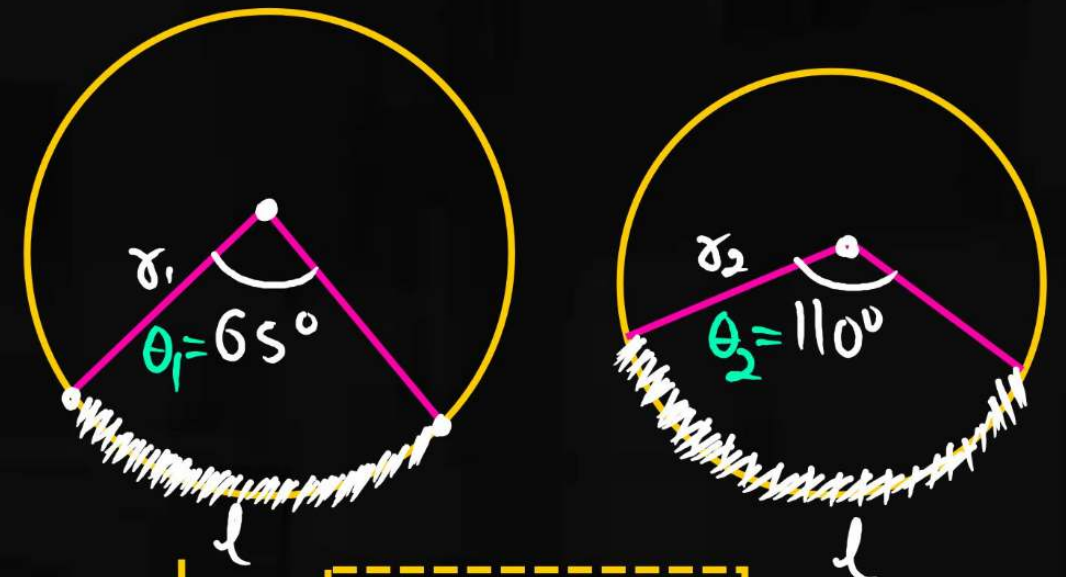
$$\Rightarrow \frac{l}{r_1} = \frac{13\pi}{36}$$

$$\Rightarrow l = \frac{13\pi}{36} r_1 \quad \text{--- (1)}$$

Also, $\theta_2 = 110^\circ = \frac{110}{180} \times \pi$

$$\Rightarrow \theta_2 = \frac{11\pi}{18} \text{ radians}$$

$$\Rightarrow \frac{l}{r_2} = \frac{11\pi}{18} \text{ radians}$$



$$l = \frac{11\pi}{18} r_2 \quad \text{--- (2)}$$

From ① and ②, we get,

$$\frac{13\cancel{\pi}}{\cancel{36}_2} r_1 = \frac{11\cancel{\pi}}{\cancel{18}_1} r_2$$

$$\Rightarrow \frac{13r_1}{2} = 11r_2$$

$$\Rightarrow \frac{r_1}{r_2} = \frac{22}{13}$$

$$\Rightarrow \boxed{r_1 : r_2 = 22 : 13}$$

So, Required Ratio = 22 : 13.

Thank
You